

Tamil Nadu - Grade 10
2018 - Mathematics

Maximum Marks: 100

1. An example for a function which is not a relation, (Domain – R, Codomain – R) is:

- (a) $y = x$
- (b) $y = x^2$
- (c) $y = x - 1$
- (d) not possible

Answer: (d)

Solution: not possible

2. If a, b, c are in G.P., then $\frac{a-b}{b-c}$ is equal to:

- (a) $\frac{a}{c}$
- (b) $\frac{a}{b}$
- (c) $\frac{c}{b}$
- (d) $\frac{b}{a}$

Answer: (b)

Solution: $\frac{a}{b}$

3. The next term of the series $\sqrt{3}, \sqrt{12}, \sqrt{27}, \dots$ is:

- (a) $\sqrt{39}$
- (b) $\sqrt{39}$
- (c) $\sqrt{54}$
- (d) $\sqrt{48}$

Answer: (d)

Solution: $\sqrt{48}$

4. What can be the degree of the remainder at most, when a fourth degree polynomial is divided by a quadratic polynomial?

- (a) 2
- (b) 0
- (c) 4
- (d) 1

Answer: (d)

Solution: 1

5. The L.C.M. of $x^3 - a^3$ and $(x - a)^2$ is:

- (a) $(x - a)^2 (x^2 + ax + a^2)$
- (b) $(x^3 - a^3) (x + a)$
- (c) $(x + a)^2 (x^2 + ax + a^2)$
- (d) $(x^3 - a^3) (x - a)^2$

Answer: (a)

Solution: $(x - a)^2 \times (x^2 + ax + a^2)$

6. $A = \begin{pmatrix} \alpha & \beta \\ \gamma & \alpha \end{pmatrix}$ is such that $A^2 = 1$, then:

- (a) $1 - \alpha^2 - \beta\gamma = 0$
- (b) $1 + \alpha^2 - \beta\gamma = 0$
- (c) $1 + \alpha^2 + \beta\gamma = 0$
- (d) $1 - \alpha^2 + \beta\gamma = 0$

Answer: (a)

Solution: $1 - \alpha^2 - \beta\gamma = 0$

7. Slope of the straight line which is perpendicular to the straight line joining the points $(-2, 6)$ and $(4, 8)$ is equal to:

- (a) -3
- (b) $\frac{1}{3}$
- (c) $-\frac{1}{3}$
- (d) 3

Answer: (a)

Solution: -3

8. The centre of a circle is $(-6, 4)$. If one end of the diameter of the circle is at $(-12, 8)$ then the other end is at:

- (a) $(-3, 2)$
- (b) $(-18, 12)$
- (c) $(0, 0)$
- (d) $(-9, 6)$

Answer: (c)

Solution: (0, 0)

9. The areas of two similar triangles are 16 cm^2 and 36 cm^2 respectively. If the altitude of the first triangle is 3 cm, then the corresponding altitude of the other triangle is:

- (a) 4 cm
- (b) 6.5 cm
- (c) 4.5 cm
- (d) 6 cm

Answer: (c)

Solution: $\sec^2 \theta - \tan^2 \theta$

10. If a vertical stick 12 m long casts a shadow 8 m long on the ground and at the same time a tower casts a shadow 40 m long on the ground, then the height of the tower is:

- (a) 75 m
- (b) 40 m
- (c) 60 m
- (d) 50 m

Answer: (c)

Solution: 60 m

11. $(1 + \cot^2 \theta)(1 - \cos \theta)(1 + \cos \theta) = \underline{\hspace{2cm}}$

- (a) $\sec^2 \theta - \tan^2 \theta$
- (b) $\tan^2 \theta - \sec^2 \theta$
- (c) $\cos^2 \theta - \sin^2 \theta$
- (d) $\sin^2 \theta - \cos^2 \theta$

Answer: (a)

Solution: $\sec^2 \theta - \tan^2 \theta$

12. If A is an acute angle of a ΔABC , right angled at B, then the value of $\sin A + \cos A$ is:

- (a) less than one
- (b) equal to one
- (c) equal to two
- (d) greater than one

Answer: (d)

Solution: greater than one

13. Radius and height of a right circular cone and that of a right circular cylinder are respectively, equal. If the volume of the cylinder is 120 cm^3 , then the volume of the cone is equal to:

- (a) 40 cm^3
- (b) 1200 cm^3
- (c) 90 cm^3
- (d) 360 cm^3

Answer: (a)

Solution: 40 cm^3

14. Standard deviation of a collection of data is $2\sqrt{2}$. If each value is multiplied by 3, then the standard deviation of the new data is:

- (a) $6\sqrt{2}$
- (b) $\sqrt{12}$
- (c) $9\sqrt{2}$
- (d) $4\sqrt{2}$

Answer: (a)

Solution: $6\sqrt{2}$

15. There are 6 defective items in a sample of 20 items. One item is drawn at random. The probability that it is a non-defective item is:

- (a) $\frac{3}{10}$
- (b) $\frac{7}{10}$
- (c) $\frac{2}{3}$
- (d) 0

Answer: (b)

Solution: $\frac{7}{10}$

16. Verify $A \subset B$ for the sets $A = \{a, b, c\}$, $B = \{1, \{a, b, c\}, 2\}$. If not justify your answer.

Solution: $A \not\subset B$ (OR) A is not a subset of B because $a \notin B$, $b \notin B$, $c \notin B$

17. If $A = \{-2, -1, 1, 2\}$ and $f = \left\{ \left(x, \frac{1}{x} \right) : x \in A \right\}$, write down the range of f . Is f a function from A to A ?

Solution: Range of $f = \{-1, 1\}$, not a function

18. Three numbers are in the ratio 2: 5: 7. If the first number, the resulting number on subtraction of 7 from the second number and the third number form an arithmetic sequence, then find the numbers.

Solution: $2x, 5x - 7, 7x$ are in A.P the required numbers are 28, 70, 98

19. In the division algorithm of polynomials the divisor is $(x + 2)$, quotient is $(x - 1)$ and the remainder is 4. Find the dividend.

Solution: Dividend = [quotient $\times$ divisor] + remainder

(OR) $(x + 2)(x - 1) + 4$

hence, dividend = $x^2 + x + 2$

20. A matrix consists of 30 elements. What are the possible orders it can have?

Solution: $1 \times 30, 2 \times 15, 3 \times 10, 5 \times 6$

$30 \times 1, 15 \times 2, 10 \times 3, 6 \times 5$

21. If $A = \begin{pmatrix} 3 & 2 \\ 4 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 0 \\ 3 & 2 \end{pmatrix}$ then find AB and BA.

Solution: $AB = \begin{pmatrix} 15 & 4 \\ 12 & 0 \end{pmatrix}$

$BA = \begin{pmatrix} 9 & 6 \\ 17 & 6 \end{pmatrix}$

22. In what ratio does the point P $(-2, 3)$ divide the line segment joining the points A $(-3, 5)$ and B $(4, -9)$ internally?

Solution: section formula(internally) $\left(\frac{lx_2 + mx_1}{l + m}, \frac{ly_2 + my_1}{l + m} \right)$ (OR) $\left(\frac{yl + 3m}{l + m}, \frac{-9l + 5m}{l + m} \right) = (-2, 3)$

$l : m = 1 : 6$

23. Find the equation of the straight line whose slope is $\frac{2}{3}$ and passing through $(5, -4)$.

Solution: Equation of straight line

(OR) $y - y_1 = m(x - x_1)$

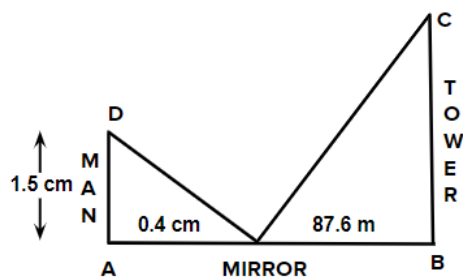
$y + 4 = \frac{2}{3}(x - 5)$

$2x - 3y - 22 = 0$

24. Draw the diagram for the given information.

A man sees the top of a tower in a mirror which is at a distance of 87.6 m. from the tower. The mirror is on the ground, facing upward. The man is 0.4 m. away from the mirror, and the distance of his eye level from the ground is 1.5 m. (The foot of man, the mirror and the foot of the tower lie along a straight line).

Solution:



25. Derive the identity $\cos^2 \theta + \sin^2 \theta = 1$ for all θ such that $0^\circ \leq \theta \leq 90^\circ$.

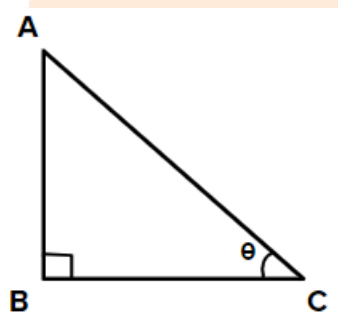
Solution: By Pythagoras theorem

$$AB^2 + BC^2 = AC^2$$

Divide by AC^2

$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$



26. Prove the identity $\sec \theta (1 - \sin \theta) (\sec \theta + \tan \theta) = 1$.

Solution: $(\sec \theta - \tan \theta)(\sec \theta + \tan \theta)$

$$= \sec^2 \theta - \tan^2 \theta$$

$$= 1 \text{ (from identity)}$$

27. A sector containing an angle of 120° is cut off from a circle of radius 21 cm and folded into a cone. Find the curved surface area of the cone. $\left(\pi = \frac{22}{7} \right)$

Solution: CSA of the cone = of sector

$$= \frac{\theta}{360} \times \pi r^2$$

$$= 462 \text{ sq. cm}$$

28. Draw the necessary table to find the standard deviation for the data 20, 14, 16, 30, 21 and 25.

Solution:

x	x^2	$x - \bar{x}$	$(x - \bar{x})^2$
14	196	-7	49
16	256	-5	25
20	400	-1	1
21	441	0	0
25	625	4	16
30	900	9	81
$\sum x = 126$	$\sum x^2 = 2818$		$\sum (x - \bar{x})^2 = 172$

$$\bar{x} = \frac{\sum x}{\sum f} = \frac{126}{6} = 21$$

$$r = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

$$= \sqrt{\frac{172}{6}} = \sqrt{28.67} = 5.354$$

29. A number is selected at random from integers 1 to 100. Find the probability that it is not a perfect cube.

Solution: $n(s) = 100$

$$P(A) = \frac{4}{100} = \frac{1}{25}$$

$$P(\bar{A}) = \frac{96}{100} = \frac{24}{25}$$

30.

(a) Solve the equation $x^2 - (\sqrt{3} + 1)x + \sqrt{3} = 0$ by completing the square method.

OR

(b) Find the total surface area of a hollow hemisphere whose outer and inner radii are 4.2 cm and 2.1 cm respectively.

Solution:

$$(a) \quad x^2 - (\sqrt{3} + 1)x + \left(\frac{\sqrt{3} + 1}{2}\right)^2 = \left(\frac{\sqrt{3} + 1}{2}\right)^2 - \sqrt{3}$$

$$x\sqrt{3} \text{ or } x = 1$$

$$(b) \text{ TSA of hollow hemisphere} = \pi (3R^2 + r^2)$$

$$= \pi (3 \times 4.2^2 + 2.1^2)$$

$$= 57.33 \pi \text{ cm}^2 \text{ or } 180.18 \text{ cm}^2$$

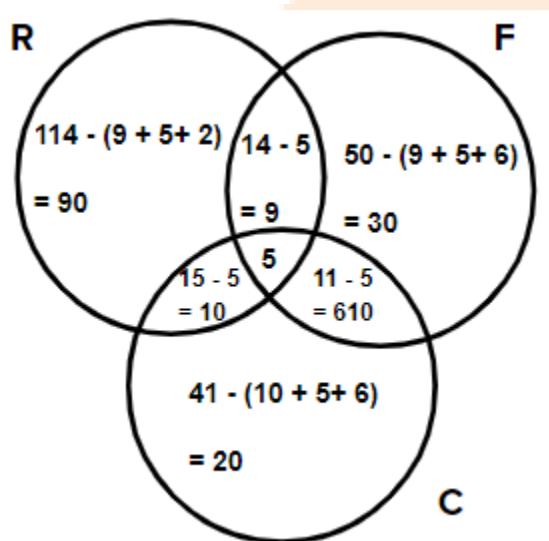
31. A radio station surveyed 190 students to determine the types of music they liked. The survey revealed that 114 liked rock music, 50 liked folk music and 41 liked classical music, 14 liked rock music and folk music, 15 liked rock music and classical music, 11 liked classical music and folk music, 5 liked all the three types of music.

Find:

(a) how many did not like any of the 3 types?

(b) how many liked any two types only?

Solution:



(a) no. of students did not like any of the 3 types = 20

(b) no. of students liked any R types only = 25

(c) no. of students liked folk music but not rock music = 36

32. A function $f : [-7, 6) \rightarrow \mathbb{R}$ is defined as follows $f(x) = \begin{cases} x^2 + 2x + 1 & ; -7 \leq x < -5 \\ x + 5 & ; -5 \leq x \leq 2 \\ x - 1 & ; 2 < x < 6 \end{cases}$

Find:

(a) $f(-7) - f(-3)$

(b) $\frac{4f(-3) + 2f(4)}{f(-6) - 3f(1)}$

Solution:

(a) $f(-7) = 36, f(-3) = 2$

$$f(-7) - f(-3) = 34$$

(b) $f(4) = 3, f(-6) = 25, f(1) = 6$

$$\frac{4f(-3) + 2f(4)}{f(-6) - 3f(1)} = \frac{14}{7} = 2$$

33. Find the three consecutive terms in an A.P. whose sum is 18 and the sum of their squares is 140.

Solution: Let the consecutive terms be $(a - d)$, a , $(a + d)$

$$(a - d) + a + (a + d) = 18$$

$$a = 6$$

$$(a - d)^2 + a^2 + (a + d)^2 = 140$$

$$d = \pm 4$$

$$2, 6, 10 \text{ (or) } 10, 6, 2$$

34. Solve $3(2x + y) = 7xy$; $3(x = 3y) = 11xy$ using elimination method.

Solution: $\frac{3}{x} + \frac{6}{y} = 7$, $\frac{a}{x} + \frac{3}{y} = 11$

$$3a + 6b = 7 ; 9a + 3b = 11$$

$$a = 1 ; b = \frac{2}{3}$$

$$x = 1, y = \frac{3}{2}$$

Solution of the system $\left(1, \frac{3}{2}\right), (0, 0) \text{ (or) } \left(1, \frac{3}{2}\right)$

35. Find the square root of the polynomial $4 + 25x^2 - 12x - 24x^3 + 16x^4$ by division method.

Solution:

$4x^2$	$4x^2 - 3x + 2$ $16x^4 - 24x^3 + 25x^2 - 12x + 4$ $16x^4$ $-24x^3 + 25x^2$ $-24x^3 + 9x^2$ $16x^2 - 12x + 4$ $16x^2 - 12x + 4$ 0
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$$\sqrt{16x^4 - 24x^3 + 25x^2 - 12x + 4} = |4x^2 - 3x + 2|$$

36. The difference of the squares of two positive numbers is 45. The square of the smaller number is four times the larger number. Find the numbers.

Solution:

$$y^2 - x^2 = 45, \quad x < y$$

$$x^2 = 4y$$

$$y^2 - 4y - 45 = 0$$

$$y = 9, \quad y = -5$$

Required number $x = 6, y = 9$

37. If $A = \begin{pmatrix} 5 & 2 \\ 7 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ verify that $(AB)^T = B^T A^T$.

Solution:

$$AB = \begin{pmatrix} 8 & -3 \\ 11 & -4 \end{pmatrix}$$

$$(AB)^T = \begin{pmatrix} 8 & 11 \\ -3 & -4 \end{pmatrix}$$

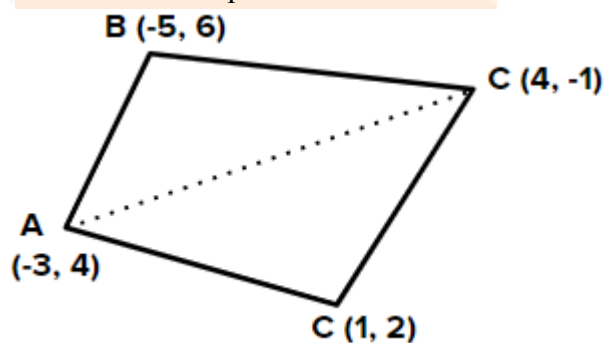
$$B^T = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$A^T = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}$$

$$B^T A^T = \begin{pmatrix} 8 & 11 \\ -3 & -4 \end{pmatrix}$$

38. Find the area of the quadrilateral whose vertices are $(-3, 4)$, $(-5, -6)$, $(4, -1)$ and $(1, 2)$.

Solution: Area of quadrilateral ABCD = ar. of $\triangle ABC$ + ar. of $\triangle ACD$



$$\text{Ar. of } \triangle = \frac{1}{2} \times [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$\text{Ar. of } \triangle ABC = \frac{1}{2} \times [-3(-6+1) + -5(-1-4) + 4(4+6)]$$

$$= \frac{1}{2} \times [15 + 25 + 40]$$

$$= 40 \text{ cm}^2$$

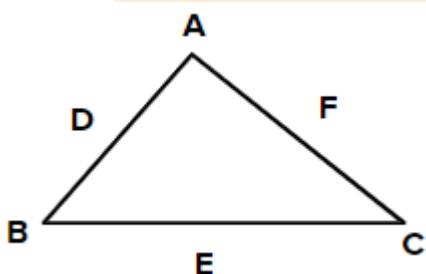
$$\text{Ar. of } \triangle ABD = \frac{1}{2} \times [-3(-1-2) + 4(2-4) + 1(4+1)]$$

$$= \frac{1}{2} \times [9 - 8 + 5]$$

$$\begin{aligned} \text{Ar. of } ABCD &= 40 + 3 \\ &= 43 \text{ cm}^2 \end{aligned}$$

39. The mid points D, E, F of the sides of a triangle ABC are (3, 4), (8, 9) and (6, 7) respectively. Find the vertices of the triangle.

Solution: Diagram mid point = $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$



$$x_1 + x_2 + x_3 = 17, y_1 + y_2 + y_3 = 20$$

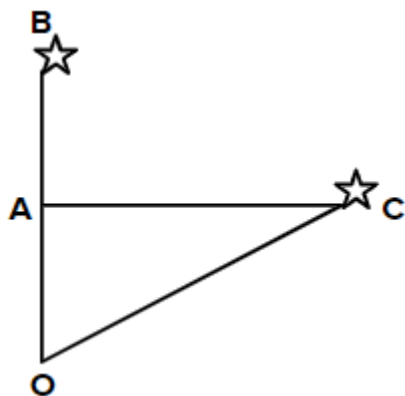
$$A(x_1, y_1) = (1, 2)$$

$$B(x_2, y_2) = (5, 6)$$

$$C(x_3, y_3) = (11, 12)$$

40. A lotus is 20 cm above the water surface in a pond and its stem is partly below the water surface. As the wind blew, the stem is pushed aside so that the lotus touched the water 40 cm away from the original position of the stem. How much of the stem was below the water surface originally?

Solution:



$$OC = OA + AB = x + 20 \text{ cm by Pythagoras theorem,}$$

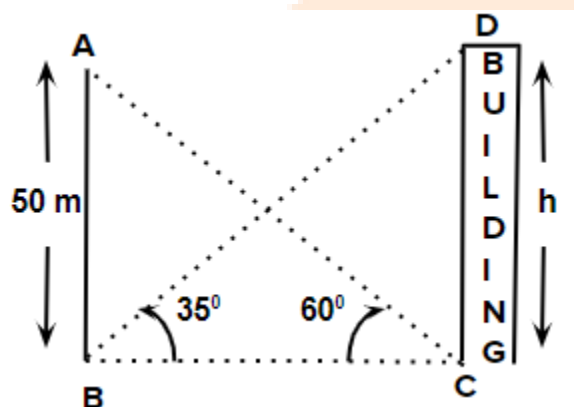
$$OC^2 = OA^2 + AC^2$$

$$(x + 20)^2 = x^2 + 40^2$$

$$x = 30 \text{ cm}$$

41. The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 50 m high, find the height of the building.

Solution:



Let BC be x

In $\triangle ABC$

$$\tan 60 = \frac{50}{x}$$

$$x = \frac{50}{\sqrt{3}}$$

In $\triangle BDC$

$$\tan 30 = \frac{h}{x}$$

$$h = \tan 30 \times x$$

$$= \frac{1}{\sqrt{3}} \times \frac{50}{\sqrt{3}} = \frac{50}{3} = 16.67 \text{ m.}$$

42. The sum of the base radius and the height of a right circular solid cylinder is 37 cm. If the total surface area of the cylinder is 1628 sq.cm, then find the volume of the cylinder.

Solution:

$$r + h = 37$$

$$2\pi h(h + r) = 1628 \text{ cm}^2$$

$$r = 7 \text{ cm}$$

$$h = 30 \text{ cm}$$

$$v = \pi r^2 h$$

$$v = 4620 \text{ cm}^3 \text{ (OR) } 1470 \pi \text{ cm}^3$$

43. For a collection of data if $\sum x = 35$, $n = 5$, $\sum (x - 9)^2 = 82$, then find $\sum x^2$ and $\sum (x - \bar{x})^2$.

Solution:

$$\bar{x} = \frac{\sum x}{n} = \frac{35}{5} = 7$$

$$\sum (x^2 - 18x + 81) = 82$$

$$\sum x^2 = 307$$

$$\sum [(x - 7) - 2]^2 = 82$$

$$\sum (x - \bar{x})^2 = 62$$

44. Two dice are rolled simultaneously. Find the probability that the sum of the numbers on the faces is neither divisible by 3 nor by 4.

Solution:

$$s = \{(1,1)(1,2)(1,3).....(6,6)\}$$

$$n(s) = 36$$

$$P(A) = \frac{12}{36}, P(B) = \frac{9}{36}$$

$$P(A \cap B) = \frac{1}{36}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{12}{36} + \frac{9}{36} - \frac{1}{36}$$

$$P(A \cup B) = \frac{20}{36}$$

$$P(A' \cap B') = \frac{4}{9}$$

45.

(a) The first term of a geometric series is 375 and the fourth term is 192. Find the common ratio and the sum of the first 14 terms.

OR

(b) Water in a cylindrical tank of diameter 4 m and height 10 m is released through a cylindrical pipe of diameter 10 cm at the rate of 2.5 km/hr. How much time will it take to empty the half of the tank? (Assume that the tank is full of water to begin with)

Solution: $t_n = ar^{n-1}$

$$(a) 375 r^3 = 192 \text{ (or) } a = 375, ar^3 = 192$$

$$r = \frac{4}{5}$$

$$s_n = a \left(\frac{1-r^n}{1-r} \right)$$

$$s_{14} = \frac{375 \left[1 - \left(\frac{4}{5} \right)^{14} \right]}{1 - \frac{4}{5}}$$

$$s_{14} = 1875 \left[1 - \left(\frac{4}{5} \right)^{14} \right]$$

(b) cylindrical tank

$$r_1 = 2\text{m}$$

$$n_1 = 10\text{m}$$

cylindrical pipe

$$r_2 = \frac{5}{100} \text{ m}, n_2 = T \times 2500 \text{ m}$$

$$\text{Volume of water discharged through cylindrical pipe} = \frac{1}{2} \{ \text{volume of water tank} \}$$

$$\pi r_2^2 h_2 = \frac{1}{2} (\pi r_1^2 h_1)$$

$$\pi \times \frac{5}{100} \times \frac{5}{100} \times T \times 2500 = \frac{1}{2} \times \pi \times 2 \times 2 \times 10$$

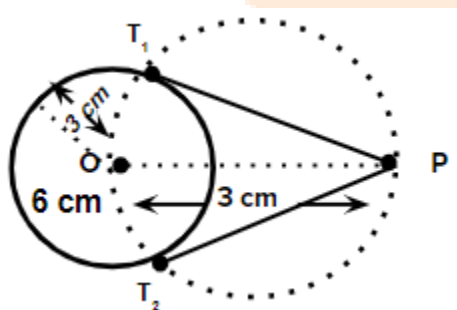
$$T = 3.2 \text{ hrs (or) } 3 \text{ hrs } 12 \text{ min}$$

46.

(a) Take a point which is 9 cm away from the centre of a circle of radius 3 cm and draw two tangents to the circle from that point and calculate their lengths.

(b) construct a cyclic quadrilateral PQRS with PQ = 4 cm, QR = 6 cm, PR = 7.5 cm, QS = 7 cm.

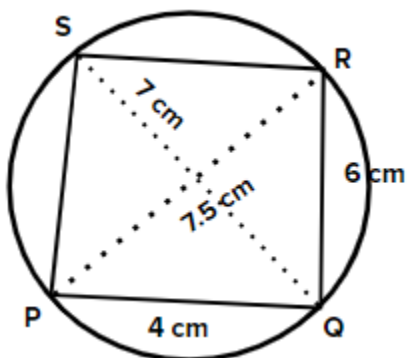
Solution:



(a) (i) Draw circle of Radius 3 cm.

- (ii) Plot point P at 9 cm from circle o.
- (iii) find mid of OP and draw circle of 4.5 cm
- (iv) wherever circle cuts the small put as T_1 and T_2
- (v) Join T_1P and T_2P to get required tangent. On measuring, $T_1P = T_2P = 8.4$ cm or 8.5 cm

(b)



- (ii) Draw $PQ = 4$ cm take arc of 7.5 from P and 6 cm from Q to get R In ΔPQR , draw $\perp$ bisectors of PQ, QR and PR to draw circumcircle
- Take $QS = 7$ cm, where ever it cuts the circumcircle plot it as s.
- Join PQRS to get cyclic quadrilateral.

47.

- (a) Draw the graph of $y = x^2 + 3x + 2$ and use it to solve the equation $x^2 + 2x + 4 = 0$.

OR

- (b) The cost of milk per litre is Rs. 15. Draw the graph for the relation between the quantity and cost. Hence find:

- (i) the proportionality constant
- (ii) the cost of 3 litres of milk.

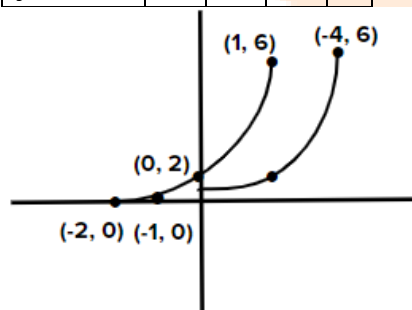
Solution:

(a)

x	-4	-3	-2	-1	0	1
y	6	2	0	0	2	6

Plot $y = x - 2$

x	-2	0	1	2
y = x - 2	-4	-2	-1	0



Plotting the points and drawing the straight line. Solution set no real roots.

(b)

x no. of tires	1	2	3	4	5	6
y cost (Rs)	15	30	45	60	75	90

